```
--- Assignmenrt => 2
create database assignment2;
use assignment2;

create table student(student_id INT PRIMARY KEY,
student_name VARCHAR(50),
city VARCHAR(30)
);

INSERT INTO student VALUES
(1, 'Amit Sharma', 'Delhi'),
(2, 'Priya Verma', 'Mumbai'),
(3, 'Ravi Kumar', 'Pune'),
(4, 'Neha Singh', 'Delhi'),
(5, 'Rahul Mehta', 'Bangalore'),
(6, 'Kiran Joshi', 'Hyderabad');

-- Q 2
CREATE TABLE courses (
course_id INT PRIMARY KEY,
course_name VARCHAR(50),
duration_months INT
);

INSERT INTO courses VALUES
(101, 'SQL Basics', 2),
(102, 'Python Programming', 3),
(103, 'Data Science', 4),
(104, 'Web Development', 5);

-- Q 3) enrollments
CREATE TABLE enrollments (
enrollment_id INT PRIMARY KEY,
student_id INT,
course_id INT,
enrollment_date DATE
);

INSERT INTO enrollments VALUES
(1001, 1, 101, '2025-06-01'),
(1002, 2, 102, '2025-06-02'),
(1003, 3, 103, '2025-06-03'),
(1004, 1, 102, '2025-06-04'),
(1005, 4, 104, '2025-06-05'),
(1006, 3, 101, '2025-06-06');

select * from enrollments;
select * from student;
select * from courses;

-- INNER JOIN (15 Questions)

-- 1. Retrieve all students with their enrolled course names.
select student.student_name,courses.course_name from student
inner join enrollments
on student.student_id = enrollments.student_id
inner join courses
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on enrollments.course_id = courses.course_id;


-- Q2. Retrieve student names, course names, and enrollment

select student.student_name,courses.course_name,enrollments.enrollment_date from student
inner join enrollments
on student.student_id = enrollments.student_id
inner join courses
on enrollments.course_id = courses.course_id;

--Q 3 . List student names and course names where the course duration is more than 2 months.

select student.student_name,courses.course_name from student
inner join enrollments
on student.student_id = enrollments.student_id
inner join courses
on enrollments.course_id = courses.course_id
where courses.duration_months> 2;

-- 4 . Retrieve all enrollments with student city.
select enrollments.enrollment_id,student.student_name,student.city,courses.course_name from
enrollments
inner join student
on enrollments.student_id = student.student_id
inner join courses
on enrollments.course_id = courses.course_id;

-- 5. Retrieve all enrollments with course duration
select student.student_name,
      courses.course_name,
      enrollments.enrollment_date,
      courses.duration_months
from student
inner join enrollments on student.student_id=enrollments.student_id
inner join courses on enrollments.course_id = courses.course_id;

-- 6. List student names enrolled in 'SQL Basics'.
select courses.course_name,
      student.student_name
from student
     inner join enrollments on student.student_id = enrollments.student_id
     inner join courses on enrollments.course_id = courses.course_id
     where course_name = "SQL Basics";

-- 7. Retrieve course names for students living in 'Delhi' --
select courses.course_name,
      student.student_name,
      student.city
from student
inner join enrollments on student.student_id = enrollments.student_id
inner join courses on enrollments.course_id = courses.course_id
where city = "delhi";

-- 8. Find students enrolled in 'Data Science' with their cities
select student.student_name,
      student.city,
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      courses.course_name
from student
    inner join enrollments on enrollments.student_id = student.student_id
    inner join courses on courses.course_id = enrollments.course_id
    where course_name = "Data Science";

-- 9. Retrieve student names, course names, and enrollment IDs
   select enrollments.enrollment_id,
            student.student_name,
            courses.course_name
     from student
      inner join enrollments on enrollments.student_id = student.student_id
      inner join courses on courses.course_id = enrollments.course_id;



-- 10. List student names with enrollment dates where course duration is less than 5 months.
              select  student.student_name,
            courses.course_id,
            enrollments.enrollment_date,
            courses.duration_months
                      from student
             inner join enrollments on enrollments.student_id = student.student_id
             inner join courses on courses.course_id = enrollments.course_id
             where duration_months <5;

-- 11. Retrieve student and course details for enrollment IDs greater than 1003.
  select student.student_name,
                        courses.course_name,
        enrollments.enrollment_id
            from student
   inner join enrollments on enrollments.student_id = student.student_id
   inner join courses on courses.course_id = enrollments.course_id
   where enrollment_id>1003;

-- 12. Retrieve all students with their courses where enrollment date is after '2025-06-02'.
select student.student_name,
      courses.course_name,
      enrollments.enrollment_date
from student
     inner join enrollments on enrollments.student_id = student.student_id
     inner join courses on courses.course_id = enrollments.course_id
where enrollment_date>"2025-06-02";

-- 13. Retrieve the student name and course name for student ID 1.
select student.student_name,
      courses.course_name,
      student.student_id
from student
     inner join enrollments on enrollments.student_id = student.student_id
     inner join courses on courses.course_id = enrollments.course_id
where student.student_id = "1";

-- 14. List the course name and student name for courses with a duration of 3 months --
select student.student_name,
      courses.course_name,
      courses.duration_months
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from student
      inner join enrollments on enrollments.student_id = student.student_id
      inner join courses on courses.course_id = enrollments.course_id
where courses.duration_months = "3";

-- 15. Retrieve enrollment details with student names for students living in 'Bangalore'.
select student.student_name,
      enrollments.enrollment_date,
      student.city
from student
    inner join enrollments on enrollments.student_id = student.student_id
    inner join courses on courses.course_id = enrollments.course_id
where student.city = "Bangalore";


-- LEFT JOIN (15 Questions)
-- 16. Retrieve all students with their enrolled courses, including students not enrolled in any course.
select student.student_name,
      courses.course_name,
      enrollments.enrollment_id
from student
left join enrollments on enrollments.student_id = student.student_id
left join courses on enrollments.course_id= courses.course_id;

-- 17. List student names, course names, and enrollment dates for all students, including those
without enrollments.
select student.student_name,
      courses.course_name,
      enrollments.enrollment_date
from student
left join enrollments on student.student_id = enrollments.student_id
left join courses on courses.course_id = enrollments.course_id;

-- 18. Find all students and their courses if enrolled; if not, show NULL for course
select student.student_name,
      courses.course_name
from student
left join enrollments on student.student_id = enrollments.student_id
left join courses on courses.course_id = enrollments.course_id;

-- 19. List all students with their course names if they are enrolled in a course with a duration of 4
months.
select student.student_name,
      courses.course_name,
      enrollments.enrollment_date
from student
left join enrollments on student.student_id = enrollments.student_id
left join courses on courses.course_id = enrollments.course_id
and courses.duration_months ='4';
--  21. List student names with their enrollment dates for all students.
select student.student_name,
      enrollments.enrollment_date
from student
            left join enrollments on enrollments.student_id = student.student_id ;

-- 22. Find students living in 'Delhi' and their courses if enrolled.
select student.student_name,
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    student.city,
    courses.course_name
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id
where student.city = 'delhi';

-- 23. Retrieve all students with course details where available.
select student.student_name,
        courses.course_id,
    courses.course_name,
    courses.duration_months
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id;

-- 24. Find all students and courses where course duration is 5 months.
select student.student_name,
    courses.duration_months
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id
where courses.duration_months = "5";

-- 25. Retrieve student names with course names for student IDs greater than 2.
select student.student_id,
        student.student_name,
    courses.course_name
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id
where student.student_id > "2";

-- 26. List all students and their enrolled course names for students living in 'Pune'.
select student.student_name,
    courses.course_name,
    student.city
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id
where student.city = 'pune';

-- 27. Retrieve all students with enrollment IDs if they are enrolled. --
select student.student_name,
    enrollments.enrollment_id
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id;

-- 28. List all students with courses they are enrolled in, even if the course name is 'Web
Development' and the student is not enrolled. --
select student.student_name,
    courses.course_name
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id
or courses.course_name = "web development";
```

```
-- 29. Retrieve all students with enrollment details for enrollment dates in June.
select student.student_name,
      enrollments.enrollment_id,
      courses.course_name,
      enrollments.enrollment_date
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id
where month(enrollments.enrollment_date) = "6" ;

-- 30. List all students and courses they are enrolled in, including students without enrollment.
select student.student_name,
      courses.course_name,
      enrollments.enrollment_date
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id;

-- RIGHT JOIN (10 Questions)
-- 31. Retrieve all courses with enrolled student names, including courses with no enrolled students.
select student.student_name,
      courses.course_name
from student
      right join enrollments on enrollments.student_id = student.student_id
      right join courses on courses.course_id = enrollments.course_id;

-- 32. List course names with student names for all courses. --
select courses.course_name,
      student.student_name
from student
     right join enrollments on enrollments.student_id = student.student_id
     right join courses on courses.course_id = enrollments.course_id;

-- 33. Retrieve all courses with student details if enrolled.
select  student.student_id,
       courses.course_name,
       student.student_name,
        student.city
from student
              right join enrollments on enrollments.student_id = student.student_id
     right join courses on courses.course_id = enrollments.course_id;

-- 34. List courses and enrolled student names where course duration is 3 months.--
select student.student_name,
      courses.course_name,
      enrollments.enrollment_date
from student
       right join enrollments on enrollments.student_id = student.student_id
       right join courses on courses.course_id = enrollments.course_id
where courses.duration_months = 3;

-- 35. Retrieve all courses with enrollment dates of students.
select courses.course_name,
      enrollments.enrollment_date
from courses
      right join enrollments on enrollments.course_id = courses.course_id;
```

```
-- 36. Find courses and student names where course ID is 101.
select courses.course_id,
      courses.course_name,
      student.student_name
from courses
            right join enrollments on enrollments.course_id = courses.course_id
     right join student on student.student_id = enrollments.student_id
where courses.course_id = 101;

-- 37. Retrieve course names and student names for enrollments made in June.
select student.student_name,
             courses.course_name,
      enrollments.enrollment_date
from student
     right join enrollments on enrollments.student_id = student.student_id
     right join courses on courses.course_id = enrollments.course_id
where month(enrollments.enrollment_date) = 6;

-- 38. List courses and student names for courses with a duration of 5 months.
select student.student_name,
      courses.course_name,
      courses.duration_months
from student
    right join enrollments on enrollments.student_id = student.student_id
    right join courses on courses.course_id = enrollments.course_id
where courses.duration_months = 5;

-- 39. Retrieve all courses and student names for students living in 'Mumbai'.
select student.student_name,
             courses.course_name,
      student.city
from student
      right join enrollments on enrollments.student_id = student.student_id
      right join courses on courses.course_id = enrollments.course_id
where student.city = "mumbai";

-- 40. Find all courses with their enrolled students or NULL if no student is enrolled.
select courses.course_name,
             student.student_name
from courses
     right join enrollments on enrollments.course_id = courses.course_id
     right join student on student.student_id = enrollments.student_id;

-- FULL JOIN (Simulated) (10 Questions)
-- 41. Retrieve all students and their enrolled courses, including students without enrollments and
courses without students.
select student.student_name,
      courses.course_name,
      enrollments.enrollment_id
from student
      left join enrollments on enrollments.student_id = student.student_id
      left join courses on courses.course_id = enrollments.course_id
union
select student.student_name,
      courses.course_name,
      enrollments.enrollment_id
```

```
from student
    right join enrollments on enrollments.student_id = student.student_id
    right join courses on courses.course_id = enrollments.course_id;

-- 42. List all students and course names using FULL JOIN simulation.
select student.student_name,
    courses.course_name
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id
union
select student.student_name,
    courses.course_name
from student
    right join enrollments on enrollments.student_id = student.student_id
    right join courses on courses.course_id = enrollments.course_id;

-- 43. Retrieve all students and course details, including unmatched data.
select courses.course_id,
    student.student_name,
    courses.course_name
from student
    left join enrollments on enrollments.student_id = student.student_id
        left join courses on courses.course_id = enrollments.course_id
union
select courses.course_id,
    student.student_name,
    courses.course_name
from student
    right join enrollments on enrollments.student_id = student.student_id
        right join courses on courses.course_id = enrollments.course_id;

-- 44. Retrieve all enrollments, student names, and course names including unmatched students and
course
select student.student_name,
    courses.course_name,
    enrollments.enrollment_id
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id
union
select student.student_name,
    courses.course_name,
    enrollments.enrollment_id
from student
    right join enrollments on enrollments.student_id = student.student_id
    right join courses on courses.course_id = enrollments.course_id;

-- 45. List student names with course names for all courses and students.
select student.student_name,
    courses.course_name
from student
    left join enrollments on enrollments.student_id = student.student_id
    left join courses on courses.course_id = enrollments.course_id
union
select student.student_name,
    courses.course_name
```

```
from student
      right join enrollments on enrollments.student_id = student.student_id
      right join courses on courses.course_id = enrollments.course_id;

-- 46. Retrieve all students with courses enrolled and courses without any students using FULL JOIN.
            select student.student_name,
          courses.course_name
            from student
        left join enrollments on enrollments.student_id = student.student_id
        left join courses on courses.course_id = enrollments.course_id
union
            select student.student_name,
          courses.course_name
            from student
        right join enrollments on enrollments.student_id = student.student_id
        right join courses on courses.course_id = enrollments.course_id;




-- 47. List all courses with enrolled students, including students without courses.
select student.student_name,
     courses.course_name
from student
     left join enrollments on enrollments.course_id = student.student_id
     left join courses on courses.course_id  = enrollments.course_id
union
select student.student_name,
           courses.course_name
from student
     right join enrollments on enrollments.student_id = student.student_id
     right join courses on courses.course_id  = enrollments.course_id;

-- 48. Retrieve all students and courses, showing NULL where there is no match.
select student.student_id,
     student.student_name,
     courses.course_name
from student
          left join enrollments on enrollments.student_id = student.student_id
          left join courses on courses.course_id = enrollments.course_id
union
         select student.student_id,
         student.student_name,
         courses.course_name
        from student
left join enrollments on enrollments.student_id = student.student_id
     left join courses on courses.course_id = enrollments.course_id;

-- 49. Find all students and courses using FULL JOIN simulation for reporting.
select student student_id,
     student.student_name,
     courses.course_name,
     courses.enrollment_date
from student
     left join enrollments on enrollments.student_id = student.student_id
     left join courses on courses.course_id = enrollments.course_id
union
```

```
 select student student_id,
      student.student_name,
      courses.course_name,
      courses.enrollment_date
from student
      right join enrollments on enrollments.student_id = student.student_id
      right join courses on courses.course_id = enrollments.course_id;

-- 50. Retrieve all students and courses, including students without enrollments and courses without
   select student.student_name,
         courses.course_name
            from student
        right join enrollments on enrollments.student_id = student.student_id
        right join courses on courses.course_id = enrollments.course_id
union
    select student.student_name,
         courses.course_name
            from student
        left join enrollments on enrollments.student_id = student.student_id
        left join courses on courses.course_id = enrollments.course_id;
```